

SAFETY QUIZ: CHE381- Chemical Engineering Laboratory

Please answer the following questions testing your knowledge of the content of the Chemical Hygiene Plan safety document. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words.

1. What is the most fundamental rule of chemical safety?

Awareness!

2. What two pieces of personal protective equipment must be worn at all times in the lab?

Safety glasses and safety shoes.

3. What additional kinds of personal protective equipment are available depending on the chemical hazard or experimental conditions?

Lab coats, splash aprons, face shields, ear protection, respirators.

4. True or False: Fume hoods are generally used as storage areas for chemicals.

False!

5. Fume hoods are to be used when the Threshold Limit Value (TLV) of a chemical is at or below:

1000 ppm or 1000 mg/M3

6. What are the four routes of chemical entry into the body?

“inhalation, ingestion, injection, and eye or skin contact”

7. True or False: Respirators are generally used in the R&D laboratories for normal work procedures.

False

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8. What is the role of the Chemical Hygiene Plan?

These measures are to ensure the protection of laboratory employees.

9. What is the guideline for laboratory room airflow and fume hood airflow?

Lab room ventilation should “not be less than 20 linear feet per minute airflow...”

Fume hoods should have “80-120 linear feet per minute airflow”

10. List 3 Common Housekeeping Practices that promote chemical hygiene and safety:

Keep pathways clear of chemicals

All work surfaces should be cleaned regularly

Access to emergency equipment should never be blocked

11. What is the proper method of transporting gas cylinders in the laboratory?

They should be transferred using a 1-cylinder, 4-wheel cart.

The cylinder cap must be on the cylinder during transportation.

Toxic, flammable, and oxidizing gases are to be stored outside in a cylinder-storage area and labeled appropriately.

12. True or False: MSDS sheets are required only for toxic chemicals and gases used in the laboratory.

False!

13. What is an accepted method of vapor detection in the laboratory?

Using mechanical vapor detectors, like a Draeger Tube.

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14. Name 3 Personal Hygiene Practices that should be followed in the laboratory.

Wash skin promptly with contact of ANY chemical

Avoid inhalation of chemicals

Do not eat or drink in chemical areas

15. Describe how individual chemicals of less than 15 lbs. (for example a bottle of acetone) are safely transported.

One could use a multiple chemical carrier with compartments for each container. The carriers should be made with heavy duty HDPE and be able to be carried with one hand.

16. **List** the Emergency Procedure to be followed for a large chemical spill or toxic gas release in the laboratory.

Evacuate the immediate area

Contact the facility operator and note the nature of emergency and location

The operator should notify all persons and notify an emergency response team

Exit the lab and deactivate lab power through emergency shut-offs

Evacuate the building and report to the department.

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17. What is the difference between a flammable chemical and a combustible chemical in terms of OSHA and NFPA guidelines.

Flammable = chemicals with a flash point below 100 F

Combustible = chemicals with a flash point between 100 and 200 F and have caused buildings to burn down.

18.  the Emergency Procedure for an individual chemical exposure in the laboratory.

Notify the facility operator and note the nature of emergency and location.

The operator may notify the fire department and/or emergency response team.

If there are no apparent dangers to yourself from exposure:

Wash contacted area for at least 15 minutes and remove all contaminated clothing.

From overexposure from inhalation, get to an area of fresh air or apply artificial respiration.

19. What is the definition of a “hazardous chemical”?

“...any chemical which can cause a physical or health a hazard.” (HCS)

20. All laboratories that handle chemicals must be equipped with eyewashes and safety showers. What is the recommended flowrate of this equipment and how often should they be checked?

The flow through fixed showers should be 30 gallons per minute and for hand-held washes, 3 gallons per minute.

Eyewashes should be checked ROUTINELY to make sure water flows through, and safety showers should be checked once a month.

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